
Exchanger type selection

Experience of a similar unit operating under similar process conditions is the best guide to exchanger type selection and should always be investigated thoroughly. However, if previous operational experience is not available, the following factors must always be considered before type selection is made.

Operating pressure and temperature

If the operating pressure is below 30 bar and the operating temperature is below 200 °C, then the plate-type exchangers must always be given serious consideration and in particular, the gasketed-plate type. At higher pressures and temperatures, the choice will be between a shell-and-tube exchanger, a gasketless plate exchanger, and a double-pipe exchanger, the latter being particularly competitive for small-capacity, high-pressure applications. If there is a cooling application an air-cooled exchanger may be considered instead of a shell-and-tube exchanger using water as coolant. For all heating duties, an electrically heated exchanger should be considered, particularly if the proposed heating medium is fouling or a suitable medium is not available.

Nature of fluids

If the corrosion characteristics of the fluids are especially aggressive, the high corrosion-resistant exchangers such as graphite, glass or Teflon™ should be considered, provided pressure, temperature and capacity are not great. If suitable plate and gasket materials are available, the gasketed-plate type should also be considered.

The lamella and spiral exchangers minimise the possibility of inter-stream leakage and for fluids which are highly fouling, or suspensions, slurries or fibrous liquids, these types merit consideration if the pressure, temperature and size are within their range. If the fluids are also corrosive the graphite, glass or Teflon™ types should be considered.

When dealing with lethal fluids, particular care must be exercised to ensure that the external joints, if any, of the selected exchanger type are of the highest integrity.

Size range

The size range of the selected type may be too small to meet the required duty without having a large number of parallel units, leading to expensive pipework and valves, with the possibility of flow distribution problems.

Cost

This is obviously an important factor but the exchanger type having the lowest capital cost may provide serious problems in service and become expensive to operate. Each case must be treated individually and it may be necessary to design and cost several alternatives before a decision is made.

Table A1.1 summarises the features of the various exchanger types.

Shell and Tube: A large number of small tubes are contained within a larger shell. The tubes are supported by baffles which provides partitioning of the shell to give cross-flow conditions in the shell side fluid.

Double Pipe: A single or multiple number of tubes inside a larger pipe (shell). This type of exchanger differs from the shell and tube in that the flow is either counter-flow or parallel-flow. For best heat transfer performance, counter-flow should be used.

Air Cooler: The best example of this type of extended surface exchangers is the radiator of a vehicle (except in this situation the air is acting as the cooling fluid).

Heat Pipe: Heat pipes rely on the evaporation and condensation of a fluid within a sealed pipe to transfer heat from one end of the pipe to the other.

Plate Heat Exchanger: The plate heat exchanger is the most common type of exchanger and is constructed from plates stacked together. The plates have patterned surfaces and the hot and cold fluid's flow between the plates. These exchangers have high heat transfer coefficients and large surface areas but are limited by pressure and temperature considerations.

Spiral Heat Exchangers: Similar to a plate exchanger except that the plates are concentric.

Lamella Exchanger: Similar to Plate exchanger: large formed metal sheets are welded together to form long flow channels or lamella (laminates). Plates are then stacked together to give the second channel. One flow flows within the plate and the other fluid flows in the space between the plates. Pure counter-flow allows reduced fouling and therefore these exchangers are used for suspensions, slurries and fibrous liquids.

Plate Fin: Fabricated from corrugated plate used on low flow rate and low capacity situations (ie gas – gas exchange).

Embossed Panel: Two metal sheets are welded together around the circumference and in predetermined locations across the surface. The plates are inflated by hydraulic pressure to give the internal flow channel. Panel can be used individually or in racks.

Graphite Heat Exchanger: The high thermal conductivity of graphite (147 W/mK), its low thermal expansion and low weight makes it desirable as a material for heat exchangers. A chemically treated graphite (it is naturally porous) can be used in shell and tube or block type exchanger designs. (Block Exchanger, a block is perforated longitudinally by horizontal rows of holes, each alternate layer being used for the process and service fluid respectively.)

Glass: Its excellent chemical resistance and smooth surface makes glass ideal for environments where non-contamination of fluids and regular sterilisation is required. Glass does however have an obvious mechanical disadvantage and a low thermal conductivity. Often used in a shell and tube arrangements.

Teflon: Extremely resistance to chemical attack, elimination of contamination of fluid and low fouling resistance. But has a poor thermal conductivity (0.19 W/mK – almost the same as a foulant!).

Rotary: The 'heat wheel' or rotary regenerator was originally designed to recover heat from the outlet from a combustion chamber (high temperature but atm pressure). A large slowly rotating wheel of coarsely knitted metal mesh absorbs heat from the hot gas stream and then delivers it to the cooler gas stream. The electrical energy required to turn the wheel is normally negligible when compared to the energy recovered.